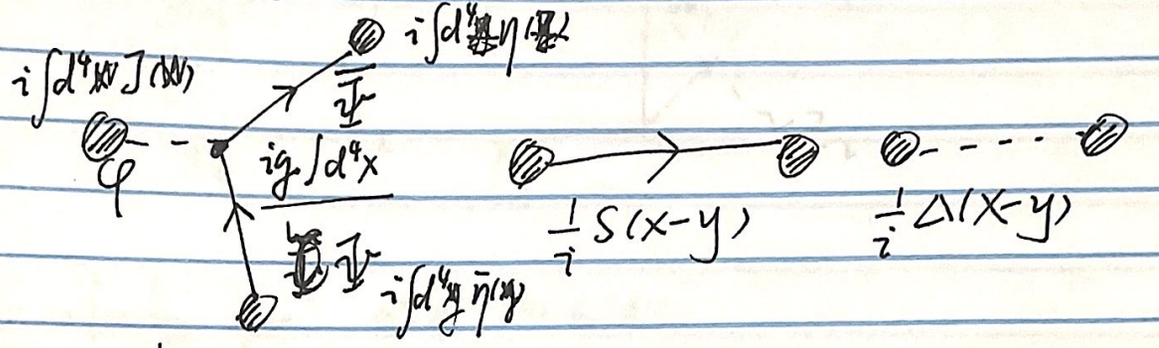
Roadmap. §45 (pp. 1–2): Yukawa correlators from LecNote17's Z, with the relative minus of fermion exchange. §§38–39 (pp. 3–13): spinors u, v at rest and boosted, $\bar{A}$, spin sums (helicity, pp. 7–8, skipped), inversion for b, d, anticommutators, Q, H; §37's Weyl field as a check (pp. 10–11). Used in LecNote19 (LSZ, §41; S from spin sums, §42) and LecNote20 ($|\mathcal{T}|^2$). Supplements: SpinorRepresentations, SymmetryFactors, DiracSpinors, FermionSigns, DiracFeynmanRules.

lec 13

Now we have three fields φ, ψ, $\bar{\psi}$

the vertex is then



For each vertex

one away

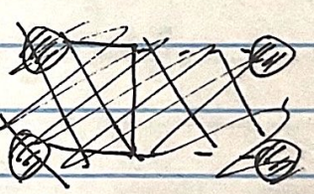
one toward.

$$\downarrow \qquad \overset{d^4w}{\downarrow}$$

$$i^3\left(\frac{1}{i}\right)^3(ig)\int d^4x\,d^4y\,d^4z\,[\bar{\eta}(x)S(x-y)S(y-z)\eta(z)]\,\Delta(y-w)J(w)$$

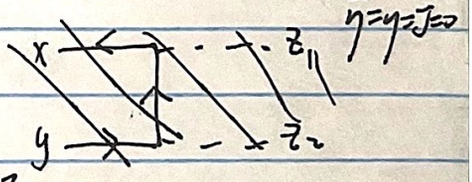
Let's say ψ is e^- and $\bar{\psi}$ is e^+

and we can calculate $e^+e^- \to \varphi\varphi$.



$$\langle 0|T\,\psi\bar{\psi}\varphi\varphi|0\rangle_c$$

$$= \frac{1}{i}\frac{\delta}{\delta\bar{\eta}_\alpha(x)}\,i\frac{\delta}{\delta\eta_\beta(y)}\,\frac{1}{i}\frac{\delta}{\delta J(z_1)}\,\frac{1}{i}\frac{\delta}{\delta J(z_2)}\,iW(\bar{\eta},\eta,J)\Big|_{\bar{\eta}=\eta=J=0}$$

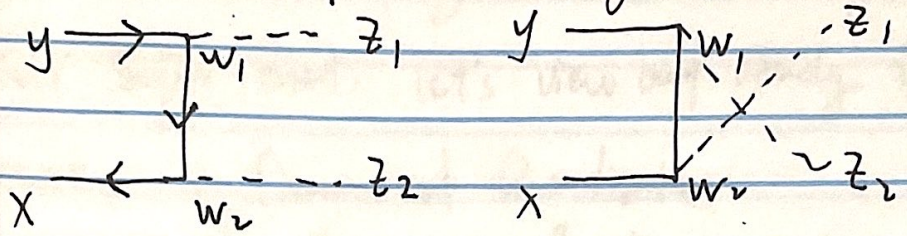


$$= \left(\frac{1}{i}\right)^5(ig)^2\int d^4w_1\,d^4w_2$$

$$[S(x-w_2)S(w_2-w_1)S(w_1-y)]_{\alpha\beta}$$

$$\Delta(z_1-w_1)\Delta(z_2-w_2) + (z_1\leftrightarrow z_2) + O(g^4)$$

Which correspond to diagrams



Similarly for $e^-e^- \to e^-e^-$

$e^-e^+ \to e^-e^+$

we need $\langle 0|T\,\psi\bar{\psi}\psi\bar{\psi}|0\rangle_c$

$$= \frac{1}{i}\frac{\delta}{\delta\bar{\eta}_{\alpha_1}(x_1)}\, i\frac{\delta}{\delta\eta_{\beta_1}(y_1)}\, \frac{1}{i}\frac{\delta}{\delta\bar{\eta}_{\alpha_2}(x_2)}\, i\frac{\delta}{\delta\eta_{\beta_2}(y_2)}$$

$$iW(\bar{\eta},\eta,J)\big|_{\bar{\eta}=\eta=J=0}$$

1 One correlator, two processes. Incoming e^- and outgoing e^+ sit at $\bar{\Psi}$ points (y), outgoing e^- and incoming e^+ at Ψ points (x): $b^\dagger \sim \int\bar{\Psi}$, $d^\dagger \sim \int\Psi$ (p. 13). For $e^-e^+ \to e^-e^+$ (e^+ in at x_2, out at y_2) the second diagram joins the incoming pair: annihilation, s-channel. Only the relative minus is physical.

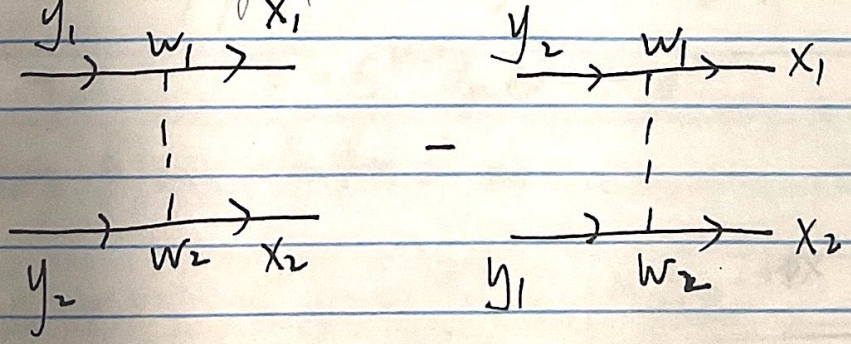
$$= \left(\frac{1}{i}\right)^5 (-ig)^2 \int d^4w_1\, d^4w_2\; [S(x_1-w_1)\,S(w_1-y_1)]_{\alpha_1\beta_1}$$

$$\Delta(w_1-w_2)\;[S(x_2-w_2)\,S(w_2-y_2)]_{\alpha_2\beta_2}$$

$$-(y_1\beta_1 \leftrightarrow y_2\beta_2) + O(g^4)$$ ①

↑ exchange of fermions

for



Next, we will "plug in" the LSZ reduction.
But before that, let's view and study $\Psi(x)$ a bit more. (Canonical Quantization
also the wave-function)
& Spinor Technology.

For the Lagrangian

$$\mathcal{L} = i\bar{\Psi}\not{\partial}\Psi - m\bar{\Psi}\Psi$$

The E.O.M. contains

$$(-i\not{\partial} + m)\Psi = 0$$

Left multiply by $(i\not{\partial} + m)$

$$(i\not{\partial}+m)(-i\not{\partial}+m)\Psi = (\not{\partial}\not{\partial} + m^2)\Psi = 0$$

$$= (-\partial^2 + m^2)\Psi = 0$$

as $\not{\partial}\not{\partial} = \gamma^\mu\gamma^\nu\partial_\mu\partial_\nu = \frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)\partial_\mu\partial_\nu$

$$= -g^{\mu\nu}\partial_\mu\partial_\nu = -\partial^2.$$

This is Klein-Gordon, so

$$\Psi(x) = u(\vec{p})\, e^{-ipx} + v(\vec{p})\, e^{-ipx}$$

$\downarrow \begin{pmatrix} \chi_a \\ \chi^{\dagger\dot{a}} \end{pmatrix}$ — 4-component spinors.

put back to the equations of motion, we get

$$(\not{p}+m)u(\vec{p})e^{-ipx} + (-\not{p}+m)v(\vec{p})e^{ipx} = 0$$

To be valid for arbitrary x and $\vec{p}$, we will need

$$(\not{p}+m)u(\vec{p}) = 0 \quad \text{(we can see next)}.$$
$$(-\not{p}+m)v(\vec{p}) = 0.$$

$u(\vec{p})$ and $v(\vec{p})$ each have two linearly independent solutions. $u_\pm(\vec{p})$, $v_\pm(\vec{p})$. Then the general solution to the Dirac Equation is

$$\psi(x) = \sum_{s=\pm} \int \widetilde{dp} \left[b_s(\vec{p}) u_s(\vec{p}) e^{-ipx} + d_s^\dagger(\vec{p}) v_s(\vec{p}) e^{ipx} \right]$$

$\widetilde{dp} \equiv \frac{d^3p}{(2\pi)^3 2w}$.

[1] Should read $\not{p} + m = m\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ (blocks of I): at rest $p_\mu = (-m, \mathbf{0})$, so $\not{p} = -m\gamma^0$. The matrix written is $-(\not{p}+m)$; same null space, so p. 5's $u_\pm(0)$ stand.

For $m \neq 0$, (any massless fermions in Nature? ✗ 1 neutrino would be)

we can go to the rest frame and have $\vec{p} = 0$, $E = m$.

Then

$$\not{p} + m = \begin{pmatrix} -m & m \\ m & -m \end{pmatrix} \text{①}, \quad -\not{p} + m = \begin{pmatrix} m & m \\ m & m \end{pmatrix}$$

and the two solutions can be separated by the spin matrix (along z)

$$S_z = \frac{i}{4}[\gamma^1, \gamma^2] = \frac{i}{2}\gamma^1\gamma^2 = \begin{pmatrix} \frac{1}{2}\sigma_3 & \\ & \frac{1}{2}\sigma_3 \end{pmatrix}$$

Real Spin

We can require

$$S_z\, u_\pm(\vec{0}) = \pm\tfrac{1}{2}\, u_\pm(\vec{0})$$

$$S_z\, v_\pm(\vec{0}) = \mp\tfrac{1}{2}\, v_\pm(\vec{0})$$

↑

Such choice conventionally leave

$$[J_z, b_\pm^\dagger(\vec{0})] = \pm\tfrac{1}{2} b_\pm^\dagger(\vec{0})$$

$$[J_z, d_\pm^\dagger(\vec{0})] = \pm\tfrac{1}{2} d_\pm^\dagger(\vec{0})$$ (1)

↑

(recall N_i, $N_i^\dagger$).

This means $b_+^\dagger(\vec{0})$ and $d_+^\dagger(\vec{0})$ each creates a particle with spin up along the z-axis.

Now, $u_+(\vec{0}) = \sqrt{m}\begin{pmatrix}1\\0\\1\\0\end{pmatrix}$, $u_-(\vec{0}) = \sqrt{m}\begin{pmatrix}0\\1\\0\\1\end{pmatrix}$

$v_+(\vec{0}) = \sqrt{m}\begin{pmatrix}0\\1\\0\\-1\end{pmatrix}$, $v_-(\vec{0}) = \sqrt{m}\begin{pmatrix}-1\\0\\1\\0\end{pmatrix}$

For the boosted spinors $\beta = \begin{pmatrix}0 & I\\ I & 0\end{pmatrix}$ (again $\beta^T = \beta^\dagger = \beta^{-1} = \beta$).

$\bar{u}_+(\vec{0}) = \sqrt{m}\,(1,0,1,0)$, $\bar{v}_+(\vec{0}) = \sqrt{m}\,(0,1,0,1)$

$\bar{u}_-(\vec{0}) = \sqrt{m}\,(0,1,0,1)$, $\bar{v}_-(\vec{0}) = \sqrt{m}\,(1,0,-1,0)$

1 Why $\mp$ for v: $[\Psi, J_z] = S_z\Psi +$ orbital (LecNote16 p. 4; the orbital part integrates away at $\mathbf{p} = 0$). $b_s^\dagger \propto \int \Psi^\dagger u_s$ keeps the sign of $S_z u_s$. $d_s^\dagger \propto \int v_s^\dagger \Psi$ (p. 13) gives $-\int (S_z v_s)^\dagger \Psi$. So a spin-up positron needs $S_z v = -\frac{1}{2} v$: a missing spin-down electron.

We can apply boost operation

$$D(\Lambda) = \exp(-i\eta\, \hat{p}\cdot K).$$

↑ unit vector in p direction

$$K^j = \frac{i}{4}[\gamma^j, \gamma^0] = \frac{i}{2}\gamma^j\gamma^0$$

$$\eta = \sinh^{-1}(|\vec{p}|/m)$$

$$u_s(\vec{p}) = e^{-i\eta \hat{p}\cdot\vec{K}}\, u_s(\vec{0})$$

$$v_s(\vec{p}) = e^{-i\eta \hat{p}\cdot K}\, v_s(\vec{0})$$

Also. $\bar{u}_s(\vec{p}) = \bar{u}_s(\vec{0})\, e^{-i\eta\hat{p}\cdot\vec{K}}$

$$\bar{v}_s(\vec{p}) = \bar{v}_s(\vec{0})\, e^{-i\eta\hat{p}\cdot\vec{K}}$$

(this follows that $\overline{K^j} = \beta K^{j\dagger}\beta = K^j$

$$= \vec{K} - \frac{i}{2}\gamma^j\gamma^{0\dagger}\gamma^{j\dagger}\gamma^0$$

$$= -\frac{i}{2}\gamma^j\gamma^0\gamma^j\gamma^0\gamma^0 = \frac{i}{2}\gamma^j\gamma^0 = K^j$$

as for general matrix $\overline{A} \equiv \beta A^\dagger \beta$. ①

So $\overline{\gamma^\mu} = \gamma^\mu$, $\overline{S^{\mu\nu}} = S^{\mu\nu}$, $\overline{i\gamma^5} = i\gamma^5$

$$\overline{\gamma^\mu\gamma^5} = \gamma^\mu\gamma^5, \quad \overline{i\gamma^5 S^{\mu\nu}} = i\gamma^5 S^{\mu\nu}$$

then we also have

$$\bar{u}_s(\vec{p})(\not{p} + m) = 0$$

$$\bar{v}_s(\vec{p})(-\not{p} + m) = 0.$$

1 Why $\bar{A} \equiv \beta A^\dagger \beta$:
$\overline{Au} = u^\dagger A^\dagger \beta = \bar{u}\bar{A}$. So barring $(\not{p}+m)u = 0$ gives $\bar{u}(\not{p}+m) = 0$, and the boost gives $\bar{u}(0)e^{-i\eta\hat{p}\cdot K}$. Also $(\bar{u}_1 A u_2)^* = \bar{u}_2 \bar{A} u_1$: $\bar{A} = A$ makes $\bar{\Psi}A\Psi$ Hermitian. Hence $i\gamma_5$ in the list, as $\bar{\gamma}_5 = -\gamma_5$. LecNote20 p. 5 uses this for $|\mathcal{T}|^2$.

Then we have some useful relations.

$$\bar{u}_{s'}(\vec{p})\,u_s(\vec{p}) = \bar{u}_{s'}(\vec{0})\,u_s(\vec{0}) = +2m\,\delta_{s's}$$

$$\bar{v}_{s'}(\vec{p})\,v_s(\vec{p}) = \bar{v}_{s'}(\vec{0})\,v_s(\vec{0}) = -2m\,\delta_{s's}$$

$$\bar{u}_{s'}(\vec{p})\,v_s(\vec{p}) = 0$$

$$\bar{v}_{s'}(\vec{p})\,u_s(\vec{p}) = 0.$$

Now study the spin sums.

$$\sum_{s=\pm} u_s(\vec{p})\,\bar{u}_s(\vec{p}) = e^{-i\eta\vec{p}\cdot\vec{K}}(-m\gamma^0+m)\,e^{-i\eta\vec{p}\cdot\vec{K}}$$

[1]

$$= -\not{p}+m$$

$$\sum_{s=\pm} v_s(\vec{p})\,\bar{v}_s(\vec{p}) = -\not{p}-m$$

1 Should read $e^{i\eta\hat{p}\cdot K}(m\gamma^0+m)e^{-i\eta\hat{p}\cdot K}$: p. 5's spinors give $\sum_s u_s(0)\bar{u}_s(0) = m(1+\gamma^0)$, which is $-\not{p}+m$ at rest; the boost sends $\gamma^0 \to -\not{p}/m$. The result is right.

The spin quantization axis is arbitrary, the above sum get rid of it and is useful for a broad class of observables.

Skip
Avoid confusing students, just mention Helicity

Helicity:

We can also get non-summed relations by projections

$$\frac{1}{2}(1+2S_z)\,u_s(\vec{0}) = \delta_{ss'}\,u_{s'}(\vec{0})$$

$$\frac{1}{2}(1-2S_z)\,v_s(\vec{0}) = \delta_{ss'}\,v_{s'}(\vec{0}).$$

Recall $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$

then $S_z = \frac{i}{2}\gamma^1\gamma^2 = -\frac{1}{2}\gamma^5\gamma^3\gamma^0$

In the rest frame, we have γ^0 as $-\not{p}/m$, γ^3 as $\not{z}$ ($z^\mu = (0, \vec{z})$)

Then $S_z = \frac{1}{2m}\gamma^5 \not{z}\not{p}$

this motivates the helicity eigenstate such that

$$u_s(\vec{p})\,\bar{u}_s(\vec{p}) \to \frac{1}{2}(1 + s\gamma^5)(-\not{p}) + \mathcal{O}(e^{-\eta})$$

$$v_s(\vec{p})\,\bar{v}_s(\vec{p}) \to \frac{1}{2}(1 - s\gamma^5)(-\not{p}) + \mathcal{O}(e^{-\eta})$$

Back to the solutions.

$$\psi = \sum_{s=\pm} \int \widetilde{dp}\,[b_s(\vec{p})u_s(\vec{p})e^{-ipx} + d_s^\dagger(\vec{p})v_s(\vec{p})e^{ipx}]$$

using the completeness relation

$$\int d^3x\, e^{ipx}\psi(x) = \sum_{s'=\pm}\left[\frac{1}{2\omega} b_{s'}(\vec{p})u_{s'}(\vec{p}) + \frac{1}{2\omega} e^{2i\omega t} d_{s'}^\dagger(-\vec{p})v_{s'}(-\vec{p})\right]$$

Given the relations above on spinors

$$b_s(\vec{p}) = \int d^3x\, e^{-ipx}\,\bar{u}_s(\vec{p})\gamma^0\psi(x)$$ [1]

$$b_s^\dagger(\vec{p}) = [b_s(\vec{p})]^\dagger = \int d^3x\, e^{ipx}\,\bar{\psi}(x)\gamma^0 u_s(\vec{p}).$$

Incorrect. I wrongly wrote $\int d^3y$ it should be $e^{-ip'x}$

1 Why $\bar{u}_s\gamma^0 = u_s^\dagger$? It must kill the $d^\dagger(-\mathbf{p})v(-\mathbf{p})$ term. K is anti-Hermitian, so $u_s(\mathbf{p})^\dagger = u_s(0)^\dagger e^{i\eta\hat{p}\cdot K}$ and $v(-\mathbf{p}) = e^{-i\eta\hat{p}\cdot K}v(0)$: $u_s^\dagger(\mathbf{p})v_{s'}(-\mathbf{p}) = u_s^\dagger(0)v_{s'}(0) = 0$ (p. 9's box; $\bar{u}$ alone fails). And $u_s^\dagger u_{s'} = 2\omega\delta_{ss'}$ cancels the $\frac{1}{2\omega}$. For d, use $v^\dagger$.

Also, we have

$$\bar{u}_{s'}(\vec{p})\,\gamma^\mu\,u_s(\vec{p}) = 2p^\mu\delta_{s's}$$

$$\bar{v}_{s'}(\vec{p})\,\gamma^\mu\,v_s(\vec{p}) = 2p^\mu\delta_{s's}$$

and $\bar{u}_{s'}(\vec{p})\,\gamma^0 v_s(-\vec{p}) = 0$

$\bar{v}_{s'}(\vec{p})\,\gamma^0 u_s(-\vec{p}) = 0$.

E.g. 38.18, 38.22. (see note (11)). used above.

$$d_s^\dagger(\vec{p}) = \int d^3x\, e^{-ipx}\,\bar{v}_s(\vec{p})\,\gamma^0\,\Psi(x)$$

$$d_s(\vec{p}) = \int d^3x\, e^{-ipx}\,\bar{\Psi}(x)\,\gamma^0\,v_s(\vec{p})$$

They have all the commutation relations of creation/annihilation operators that we want.

Here we

$$b_s(\vec{p})|0\rangle = d_s(\vec{p})|0\rangle = 0.$$

Recall

$$Q = \int d^3x\,\bar{\Psi}\gamma^0\Psi$$

$$= \sum_{s=\pm}\int \widehat{dp}\,\left[b_s^\dagger(\vec{p})\,b_s(\vec{p}) + d_s(\vec{p})\,d_s^\dagger(\vec{p})\right]$$

$$= \sum_{s=\pm}\int \widehat{dp}\,\underbrace{\left[b_s^\dagger(\vec{p})\,b_s(\vec{p})\right]}_{\#\text{ of } b\text{-type}} - \underbrace{d_s^\dagger(\vec{p})\,d_s(\vec{p})}_{\#\text{ of } d\text{-type particles.}} + \text{const}$$

Note that what we have done is corresponding to Canonical quantization, which means

$$\{\bar{\Psi}_\alpha(\vec{x},t), \bar{\Psi}_\beta(\vec{y},t)\} = 0$$

$$\{\bar{\Psi}_\alpha(\vec{x},t), \bar{\Psi}_\beta(\vec{y},t)\} = (\gamma^0)_{\alpha\beta}\,\delta^3(\vec{x}-\vec{y})$$

α, β are just which element $\alpha, \beta = 1, 2, 3, 4$.

Recall γ^0 is a 4×4 matrix

Alternatively, one can also find this is consistent with per Wyle spinor quantization

e.g. For general one wyle field Lagrangian density

$$\mathcal{L} = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \frac{1}{2}m(\psi\psi + \psi^\dagger\psi^\dagger)$$

conjugate momentum is then

$$\Pi^a(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_a(x))}$$

$$\mathcal{H} = -i\psi^\dagger \bar{\sigma}^i \partial_i \psi + \frac{1}{2}(\psi\psi + \psi^\dagger\psi^\dagger)$$

Fermions are anti-commuting, so we have

$$\{\psi_a(\vec{x},t), \psi_b(\vec{y},t)\} = 0$$

$$\{\psi_a(\vec{x},t), \Pi^c(\vec{y},t)\} = i\delta_a{}^c\,\delta^3(\vec{x}-\vec{y})$$

With substitution, we get

$$\{\psi_a(\vec{x},t), \psi_i^\dagger(\vec{y},t)\}\,\bar{\sigma}^0{}^{\dot{c}a} = \delta_a{}^c\,\delta^3(\vec{x}-\vec{y})$$

Noting that $\bar{\sigma}^0 = \mathbb{1}_{2\times 2}$, $\sigma = \mathbb{1}_{2\times 2}$

$$\{\psi_a(\vec{x},t), \psi_i^\dagger(\vec{y},t)\} = \sigma^0_{a\dot{c}}\,\delta^3(\vec{x}-\vec{y})$$

$$\{\psi^a_\alpha(\vec{x},t), \psi^{\dot{c}\dagger}(\vec{y},t)\} = \bar{\sigma}^{0\dot{c}a}\,\delta^3(\vec{x}-\vec{y}).$$

Eq. 33.13, Gordon Identity

$$\gamma^\mu \not{p} = \frac{1}{2}\{\gamma^\mu, \not{p}\} + \frac{1}{2}[\gamma^\mu, \not{p}]$$

$$= -p^\mu - 2i S^{\mu\nu} p_\nu$$

and $\not{p}'\gamma^\mu = -p'^\mu + 2i S^{\mu\nu} p'_\nu$. (1)

1 Should read $\not{p}'\gamma^\mu = -p'^\mu + 2iS^{\mu\nu}p'_\nu$. With the line above, the identity named here is $2m\bar{u}'\gamma^\mu u = \bar{u}'[(p+p')^\mu - 2iS^{\mu\nu}(p'-p)_\nu]u$: a convection term plus a spin term.

$$2m\bar{u}_{s'}(\vec{p}')\gamma^\mu u_s(\vec{p}) = -\bar{u}_{s'}(\vec{p}')(\not{p}'\gamma^\mu + \gamma^\mu\not{p})u_s(\vec{p}).$$

Similarly,

$$2m\bar{v}_{s'}(\vec{p}')\gamma^\mu v_s(\vec{p}) = -\bar{v}_{s'}(\vec{p}')(\not{p}'\gamma^\mu + \gamma^\mu\not{p})v_s(\vec{p})$$

when $p = p'$

$$\bar{u}_s(\vec{p})\gamma^\mu u_s(\vec{p}) = 2p^\mu \delta_{ss'}$$

$$\bar{v}_{s'}(\vec{p}')\gamma^\mu v_s(\vec{p}') = 2p^\mu \delta_{ss'}$$

With $\vec{p}' = -\vec{p}$, $p'^0 = p^0$

$$\bar{u}_{s'}(\vec{p}')\gamma^0 u_s(\vec{p}) = 0$$

$$\bar{v}_{s'}(\vec{p}')\gamma^0 v_s(-\vec{p}) = 0$$

1 Should read $\bar{u}_{s'}(\mathbf{p})\gamma^0 v_s(-\mathbf{p}) = 0 = \bar{v}_{s'}(\mathbf{p})\gamma^0 u_s(-\mathbf{p})$, as boxed on p. 9 (proof: p. 8 note). They give $\{b, d\} = 0$ and kill the cross terms in Q, H. The u–u, v–v versions equal $2m\delta_{s's}$.

The full list of anti-commutation relations on the solutions we got are (momentum space)

$$\{b_s(\vec{p}), b_{s'}(\vec{p}')\} = 0$$

$$\{d_s(\vec{p}), d_{s'}(\vec{p}')\} = 0 \quad \Leftarrow \quad \{-\Pi_\alpha(\vec{x},t), -\Pi_\beta(\vec{y},t)\} = 0$$

$$\{b_s(\vec{p}), d^\dagger_{s'}(\vec{p}')\} = 0$$

and also the Hermitian conjugated relations are true.

Now we also have non-trivial ones.

$$\{b_s(\vec{p}), b^\dagger_{s'}(\vec{p}')\} = \int d^3x\, d^3y\, e^{-ipx+ip'y}\, \bar{u}_s(\vec{p})\gamma^0 \{-\Pi(x), -\Pi(y)\}\gamma^0 u_{s'}(\vec{p}')$$

$$= \int d^3x d^3y\, e^{-ipx+ip'y}\, \bar{u}_s(\vec{p})\gamma^0\gamma^0\gamma^0\, \delta^3(\vec{x}-\vec{y})\, u_{s'}(\vec{p}')$$

$$= (2\pi)^3 \delta^3(\vec{p}-\vec{p}')\, \bar{u}_s(\vec{p})\gamma^0 u_{s'}(\vec{p}')$$

$$= (2\pi)^3 \delta^3(\vec{p}-\vec{p}')\, 2\omega\, \delta_{ss'}$$

Similarly

$$\{d_s^\dagger(\vec{p}),\, d_{s'}(\vec{p}')\} = (2\pi)^3 \delta^3(\vec{p}-\vec{p}')\, 2\omega\, \delta_{ss'}$$

And in momentum space

$$H = \int d^3x\, \bar{\psi}(-i\gamma^0\partial_i + m)\psi$$

$$= \sum_{s=\pm} \int \widetilde{dp}\; \omega \left[b_s^\dagger(\vec{p}) b_s(\vec{p}) + d_s^\dagger(\vec{p}) d_s(\vec{p}) \right] - 4\mathcal{E}_0 V$$

local | sho?

1 Why anticommutators? The v-modes carry $-\omega$, $(-\gamma^i p_i + m)v = -\omega\gamma^0 v$, so $H = \sum\int \widetilde{dp}\,\omega[b^\dagger b - dd^\dagger]$. With $\{d, d^\dagger\}$: $-dd^\dagger = d^\dagger d -$const, bounded below. With commutators each positron lowers H: no vacuum. Likewise Q (p. 9) becomes $N_b - N_d$. Constant: four oscillators per $\mathbf{p}$ (b, d, two spins), $-\frac{1}{2}\omega$ each.

Now LSZ reduction.

Let's denote

$$|p, s, +\rangle = b_s^\dagger(\vec{p})|0\rangle$$

$$|p, s, -\rangle = d_s^\dagger(\vec{p})|0\rangle$$

"$\pm$" to denote particle type $b_s^\dagger$ and $d_s^\dagger$ (which have $u(1)$ charge oppositely) and as

$$b_s^\dagger(\vec{p}) = \int d^3x\, e^{ipx}\, \bar{\psi}(x)\gamma^0 u_s(\vec{p})$$

$$d_s^\dagger(\vec{p}') = \int d^3x\, e^{ipx}\, \bar{v}_s(\vec{p})\gamma^0 \psi(x)$$

← for free theory.

We have

$$\boxed{\langle p, s, q | p', s', q' \rangle = (2\pi)^3 2\omega\, \delta^3(\vec{p}-\vec{p}')\, \delta_{ss'}\delta_{qq'}}$$